

Overall Algorithm Documentation

Mobile and Web Payroll / POS Management System

Scope: POS sales and inventory, quotations, payroll, attendance/DTR, owner requisitions, damage items, and loan deductions.

General conclusion: The system uses several practical algorithms. It does not use one single algorithm for every module. Most modules use formulas, aggregation, validation, sorting, grouping, and database transactions. FIFO is used only for processing loan deductions in date order.

1. Algorithm Classification Overview

Module	Main algorithm type	FIFO?
POS sales	Linear aggregation + arithmetic validation	No
Inventory	Direct quantity update with safeguards	No
Quotation	Rule-based formula pricing + aggregation	No
Payroll	Grouping + payroll formula computation	Partly: loans
Attendance / DTR	Chronological sorting + time-difference calculation	No
Owner requisitions	Arithmetic + state-machine workflow	No
Damage items	Transactional inventory adjustment	No
Loan deductions	Oldest-first ordered processing	Yes, FIFO-like

2. POS Sales Algorithm

Location: Mobile POS sale screen and the shared PHP backend.

- Read the cart items, product prices, quantities, discount, and payment.
- Compute each line total using price multiplied by quantity.
- Aggregate all line totals to get the subtotal.
- Clamp negative discounts to zero and cap discounts at the subtotal.
- Compute the final sale total after discount.
- Compute change only when payment is greater than zero.
- Save the sale header and line items inside a database transaction.

```
line_total = price x quantity
```

```
subtotal = sum(line_total) total = subtotal - discount change = max(0, payment - total)
```

Complexity: $O(n)$, where n is the number of cart items. **FIFO:** No.

3. Inventory Algorithm

- Adding stock directly increases quantity_on_hand.
- Selling products directly decreases quantity_on_hand.
- The sale and damage logic prevents stock from going below zero using a maximum safeguard.
- The system does not store inventory batches, purchase dates, or per-batch costs.

```
new_stock = current_stock + added_quantity sale_stock = max(current_stock - sold_quantity, 0)
```

Algorithm type: direct quantity update. **FIFO:** No inventory FIFO costing is implemented.

4. Quotation Algorithm

Mobile calculation: quotationCalculator.ts and POSQuotationTab.tsx. **Backend validation:** admin_class.php recalculates the saved quotation totals.

4.1 Dimensions and materials

- Convert the selected measurement unit into meters or square meters.
- Calculate glass area from width and height.
- Calculate perimeter for aluminum requirements.
- Apply thickness and glass-color multipliers.
- Add design, accessories, delivery/installation, and measurement fees.

```
area = width x height x unit_conversion perimeter = 2 x (width + height) x unit_conversion
glass_cost = base_price x area x thickness_multiplier x color_multiplier x panels
aluminum_cost = base_price x 0.5 x perimeter x panels
```

4.2 Totals, discount, and tax

```
subtotal = sum(all line totals) discount = fixed discount or percentage discount, capped at
subtotal tax = (subtotal - discount) x tax_percentage / 100 total = subtotal - discount +
tax
```

Algorithm type: deterministic rule-based pricing plus linear aggregation. **FIFO:** No.

5. Payroll Algorithm

Location: Web/backend payroll calculation. Mobile mainly records and synchronizes DTR data.

- Select approved DTR records within the payroll date range and assigned branches.
- Group records by employee.
- Cap regular work hours at 8 hours per day.
- Convert hours to present-day equivalents, including the 4.5625-hour half-day rule.
- Calculate salary from daily rate and worked hours.
- Accumulate overtime, late minutes, undertime, allowances, contributions, deductions, and loans.
- Save payroll items in a transaction and update payroll status.

```
hourly_rate = daily_salary / 8 worked_salary = work_hours x hourly_rate per_minute_rate =
daily_salary / 480 late_deduction = late_minutes x per_minute_rate
```

Algorithm type: grouping and multi-component payroll computation. **FIFO:** Only loan processing is FIFO-like.

6. Attendance / DTR Algorithm

- Attendance logs are sorted by date/time.
- The first and last relevant logs are used to determine the work duration.
- A one-hour break is subtracted in the mobile hours calculation.
- Hours are grouped by employee and date for reporting and payroll.
- Late, overtime, undertime, and attendance type are retained for payroll processing.

```
worked_hours = last_log_time - first_log_time - break_time if worked_hours < 0, use 0
```

Algorithm type: chronological sorting and time-difference calculation. **FIFO:** No.

7. Owner Requisition Algorithm

- Validate item name and positive quantity.
- Calculate the expected total from quantity and unit price on mobile.
- Create a requisition with Pending status.
- Allow status transitions from Pending to Approved or Rejected.
- Record payments and compute the remaining balance on mobile.
- Use local-first storage on mobile, then attempt server synchronization.

```
total_price = quantity x unit_price balance = amount_paid - total_price
```

Algorithm type: arithmetic plus a state-machine workflow. **FIFO:** No.

8. Damage Item Algorithm

- Validate product/item name, branch, cashier, and positive quantity.
- If linked to a product, deduct the damaged quantity from inventory.
- Insert the damage record with Pending status.
- Perform inventory deduction and damage insertion in one transaction.
- When a damage record is deleted, restore the quantity to inventory and then delete the record.
- Allow approval only for records still in Pending status.

```
remaining_stock = max(current_stock - damaged_quantity, 0) restored_stock = current_stock + damaged_quantity
```

Algorithm type: transaction-based inventory adjustment and status validation. **FIFO:** No.

9. Loan Deduction Algorithm

- Select active loans with a remaining balance.
- Sort loans by loan_date ascending, then loan_id ascending.
- Process the oldest loan first.
- Limit each deduction to the remaining loan balance.
- Continue through the ordered list of loans.

```
deduction_amount = min(requested_deduction, remaining_balance)
```

Algorithm type: FIFO / oldest-first ordered processing. This is the clearest FIFO implementation in the system.

10. Final Conclusion

The system uses multiple algorithms rather than one universal algorithm. POS and quotation use arithmetic and aggregation. Attendance uses sorting and time calculations. Payroll uses grouping and business rules. Requisitions use workflow states. Damage uses database transactions. Loan deductions use FIFO-like oldest-first ordering.

Overall classification: business-rule algorithms, linear scans, aggregation/reduction, chronological sorting, state-machine validation, and transactional database processing.

Generated from the current mobile and payroll codebase.